Post-operative oxygen delivery and delirium after cardiac surgery with cardiopulmonary bypass: a multisite retrospective cohort study of 1,690 patients

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Research Problem

Motivation: Delirium is a common, distressing, and dangerous complication after cardiac surgery with cardiopulmonary bypass (CPB) with few modifiable risk factors.

Delirium is associated with significant morbidity, long term cognitive decline, and lower quality of life after discharge.

Key question: We evaluated the association between post-operative oxygen delivery (DO_2i) and the risk of delirium and identify a critical threshold for DO_2i associated with delirium risk after surgery.

Background and Prior Work

Pharmacologic interventions such as dexmedetomidine, ketamine, methylprednisolone, melatonin, and ramelteon demonstrated little or no benefit.

Non-pharmacological interventions such as remote ischemic preconditioning, EEG guided anesthetic depth, cerebral oximetry, and limiting hyperoxia have also been ineffective.

Intraoperative DO_2i: Leenders et al. reported that lower oxygen delivery during CPB was associated with delirium after surgery with an indexed oxygen delivery (DO_2i) critical threshold for predicting delirium of 310 ml/min/m^2.

Limitations: While intraoperative DO_2i has been implicated in developing delirium, it is unknown if DO_2i during the postoperative recovery in the cardiothoracic intensive care unit (CTICU) is associated with the incidence of delirium.

Key Results

The **185 patients (11%)** who developed delirium had a **lower postoperative DO_2i** (mean difference 26 ml/min/m^2) compared to the 1,505 patients who did not develop delirium.

Multivariable analysis controlling for known risk factors showed post-operative mean DO_2i was not independently associated with delirium.

Post-operative DO2i was **inversely correlated** with CPB time with a decrease of **0.27 ml/min/m^2** per minute of CPB exposure and statistically significant.

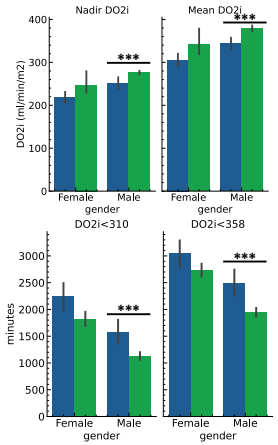
The critical threshold of DO_2i after cardiac surgery is **lower** with increased **age** and **female sex**.

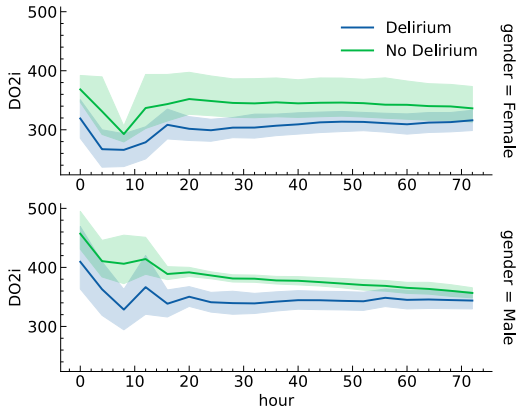
Birth-gender-stratified exploratory analysis identified an **independent association** between **delirium** and **decreased mean DO_2i** between 8 and 16 hours and 24 and 32 hours in **male patients**.

Technical Details

1) We analyzed data from **1690 patients** (1217 male, 473 female) who underwent elective cardiac surgery **with CPB** between 2011 and 2022 at two centers.
2) CAM-ICU scores were documented every 12 hours after CTICU admission and **delirium** occurred in 185 patients (11%) within the first 7 post-operative days.
3) We calculated time-varying **oxygen content (CaO_2)** for each patient using measurements of hemoglobin (Hgb) concentration, arterial saturation (SpO_2) and oxygen tension (PaO_2). Using **continuous cardiac output (CO)** from indwelling pulmonary artery catheters we calculated time-varying **indexed oxygen delivery (DO_2i)** for each patient.

Figures





Discussion

Optimizing DO2i after cardiac surgery may be a viable therapeutic strategy to mitigate the increased risk of delirium due to advanced age and prolonged CBP exposure. Goal-directed perfusion practices during CBP are effective for reducing acute kidney injury after cardiac surgery and are recommended by the recent clinical guidelines for perioperative care in cardiac surgery and supported by multiple studies. Like our study, Leenders et al. observed that lower oxygen delivery during CPB is associated with delirium after surgery although only age, peripheral vascular disease, chronic kidney disease, and cross-clamp time were independently associated with delirium. We observed a DO2i critical threshold 358 ml/min/m^2. Further studies are necessary to determine whether targeting a higher DO2i in older patients who require a long CPB time reduces the incidence of delirium.

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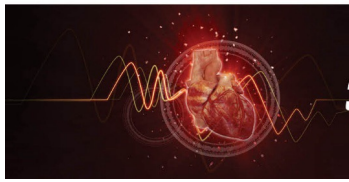

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